

LINEAR ALGEBRA

Fall 2026

Week 1

About the Independent Study Assignments

The Independent Study Assignments are homework problems and assignments that will help you learn the week's content more deeply by solving problems and expanding your knowledge and skills. *Your answers will not be submitted and will not be graded.* You may solve the problems individually or collaborate with your fellow students or ask the assistance from an AI tool (to help you clarify the concepts, not providing the solution without you thinking about it first).

Independent Study Assignments (Homework Problems): Vectors

These exercises emphasize the textbook's central ideas of vector addition, scalar multiplication, linear combinations, span and linear independence. These exercises do *not* require dot products, cross products or matrices.

Unless otherwise stated, vectors are column vectors with real components.

Part A — Basic vector operations

Exercise 1: Vector components

Let

$$\mathbf{u} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} -1 \\ 5 \end{bmatrix}.$$

Calculate:

1. $\mathbf{u} + \mathbf{v}$;
 2. $\mathbf{u} - \mathbf{v}$;
 3. $2\mathbf{u}$;
 4. $-3\mathbf{v}$;
 5. $2\mathbf{u} - 3\mathbf{v}$.
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Exercise 2: Three-dimensional vectors

Given

$$\mathbf{a} = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 3 \\ 2 \\ -2 \end{bmatrix},$$

find:

$$\mathbf{a} + \mathbf{b}, \quad \mathbf{a} - \mathbf{b}, \quad 3\mathbf{a} + 2\mathbf{b}.$$

Exercise 3: Find the unknown vector

Find $\mathbf{x}$ if

$$\mathbf{x} + \begin{bmatrix} 2 \\ -4 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \\ 3 \\ -2 \end{bmatrix}.$$

Exercise 4: Vector equation

Find $\mathbf{x}$ if

$$3\mathbf{x} - 2\mathbf{u} = \mathbf{v},$$

where

$$\mathbf{u} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 8 \\ 2 \end{bmatrix}.$$

Part B — Length and direction

Exercise 5: Vector length

Calculate the Euclidean length of each vector:

$$\mathbf{u} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 2 \\ -3 \\ 6 \end{bmatrix}, \quad \mathbf{w} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Recall that

$$\|\mathbf{x}\| = \sqrt{x_1^2 + \cdots + x_n^2}.$$

Exercise 6: Unit vector

Find a unit vector in the same direction as

$$\mathbf{v} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}.$$

Verify directly that your answer has length 1.

Exercise 7: Vector with prescribed length

Find a vector of length 10 pointing in the same direction as

$$\mathbf{v} = \begin{bmatrix} 3 \\ -4 \end{bmatrix}.$$

Exercise 8: Distance between points

Using vector subtraction and vector length, calculate the distance between

$$P = (1, -2, 3)$$

and

$$Q = (5, 1, -3).$$

Part C — Linear combinations

Exercise 9: Compute linear combinations

Let

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 3 \\ -1 \end{bmatrix}.$$

Calculate:

1. $2\mathbf{v}_1 + \mathbf{v}_2$;
 2. $-\mathbf{v}_1 + 3\mathbf{v}_2$;
 3. $4\mathbf{v}_1 - 2\mathbf{v}_2$;
 4. $0\mathbf{v}_1 + 5\mathbf{v}_2$.
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Exercise 10: Find the coefficients (weights)

Determine c_1 and c_2 such that

$$c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 7 \\ 1 \end{bmatrix}.$$

Verify your answer by substitution.

Exercise 11: Three-vector combination

Determine whether

$$\mathbf{b} = \begin{bmatrix} 5 \\ 1 \\ 4 \end{bmatrix}$$

can be written in the form

$$\mathbf{b} = c_1 \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

If possible, find c_1, c_2, c_3 .

Part D — Span and linear independence

Exercise 12: Span of one vector

Describe geometrically all vectors of the form

$$c \begin{bmatrix} 2 \\ -3 \end{bmatrix}, \quad c \in \mathbb{R}.$$

Determine whether each vector belongs to this span:

$$\begin{bmatrix} 4 \\ -6 \end{bmatrix}, \quad \begin{bmatrix} -2 \\ 3 \end{bmatrix}, \quad \begin{bmatrix} 6 \\ -8 \end{bmatrix}.$$

Exercise 13: Linear dependence

Determine whether the following vectors are linearly independent:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 3 \\ 6 \end{bmatrix}.$$

Find nonzero numbers c_1, c_2 satisfying

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 = \mathbf{0}.$$

Explain the result geometrically.

Exercise 14: Independence in $\mathbb{R}^3$

Determine whether

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

are linearly independent.

If they are dependent, express one vector as a linear combination of the other two.

Additional exercises on span

Exercise 15: Do two vectors span $\mathbb{R}^2$?

Consider

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

1. Determine whether $\mathbf{v}_1$ and $\mathbf{v}_2$ span $\mathbb{R}^2$.
2. Given an arbitrary vector

$$\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix},$$

find coefficients c_1, c_2 such that

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 = \mathbf{b}.$$

3. Use your formula to express

$$\begin{bmatrix} 7 \\ 4 \end{bmatrix}$$

as a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$.

Exercise 16: Vectors that do not span $\mathbb{R}^2$

Consider

$$\mathbf{u}_1 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \quad \mathbf{u}_2 = \begin{bmatrix} -6 \\ 3 \end{bmatrix}.$$

1. Show that $\mathbf{u}_2$ is a scalar multiple of $\mathbf{u}_1$.
2. Explain why the two vectors do not span $\mathbb{R}^2$.
3. Describe geometrically the set

$$\text{span}\{\mathbf{u}_1, \mathbf{u}_2\}.$$

4. Determine whether each vector belongs to this span:

$$\begin{bmatrix} 8 \\ -4 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Exercise 17: Do three vectors span $\mathbb{R}^3$?

Let

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

1. Given an arbitrary vector

$$\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix},$$

solve

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + c_3 \mathbf{v}_3 = \mathbf{b}$$

for c_1, c_2, c_3 .

2. Use your result to explain why the three vectors span $\mathbb{R}^3$.
3. Express

$$\begin{bmatrix} 4 \\ 2 \\ 6 \end{bmatrix}$$

as a linear combination of the three vectors.

Part E — Engineering applications

Exercise 18: Force balance

Three forces act on a mechanical component:

$$\mathbf{F}_1 = \begin{bmatrix} 40 \\ 10 \end{bmatrix} \text{ N}, \quad \mathbf{F}_2 = \begin{bmatrix} -15 \\ 25 \end{bmatrix} \text{ N}, \quad \mathbf{F}_3 = \begin{bmatrix} -5 \\ -20 \end{bmatrix} \text{ N}.$$

1. Find the resultant force.
2. Find a fourth force $\mathbf{F}_4$ that would place the component in equilibrium:

$$\mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 + \mathbf{F}_4 = \mathbf{0}.$$

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